

Version number: Ver 0.1.8

# ESP32-S3 Mini Module Product Specifications

2026/07/27



## Change Log(Reversion Record)

| Date<br>(Date) | Version<br>(Rev) | Notes<br>(Change Description)                                                                                                                                                                                                                            | Author<br>(Author) |
|----------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2025-12-17     | Ver0.1.0         | Draft Product Specifications                                                                                                                                                                                                                             | Wu Chaoqun         |
| 2025-1-19      | Ver0.1.1         | Update the content in the 'How to Use', 'System Requirements', 'Frequently Asked Questions' and 'About Us' sections                                                                                                                                      | GongRui            |
| 2026-4-1       | Ver0.1.2         | Update product images                                                                                                                                                                                                                                    | GongRui            |
| 2026-5-27      | Ver0.1.3         | Update 3D renderings                                                                                                                                                                                                                                     | GongRui            |
| 2026-6-10      | Ver0.1.4         | Update application scenarios                                                                                                                                                                                                                             | GongRui            |
| 2026-6-10      | Ver0.1.5         | In accordance with FCC KDB 996369 D03 specifications, new sections have been added covering RF compliance, antenna integration, RF exposure, testing requirements, disclaimers and change management, thereby refining the OEM integration requirements. | GongRui            |
| 2026-6-15      | Ver0.1.6         | Update product images                                                                                                                                                                                                                                    | GongRui            |
| 2026-6-26      | Ver0.1.7         | Update Authentication Authority Requirements                                                                                                                                                                                                             | GongRui            |
| 2026-7-27      | Ver0.1.8         | Update Authentication Requirements                                                                                                                                                                                                                       | GongRui            |

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# 1 Product Overview

## 1.1 Product name

### ESP32-S3 Mini Module

## 1.2 Product Overview

The ESP32-S3 Mini Module is a high-performance, ultra-compact embedded development module designed specifically for IoT applications, wearable devices and small smart monitoring devices. It is powered by a high-performance ESP32-S3 LX7 dual-core 32-bit processor with a maximum clock speed of 240MHz, and features an integrated neural network accelerator capable of supporting basic AI dialogue and inference tasks.

This module offers extensive wireless connectivity, supporting Wi-Fi (802.11b/g/n) . It provides 32 GPIO pins and a variety of communication interfaces, with optional versions featuring on-board temperature and inertial sensors; Measuring 24.54 mm × 17.78 mm × 4.50 mm and weighing 2.44 g, it is compatible with development environments such as ESP-IDF and Arduino IDE, comes with comprehensive development resources, and is suitable for space-constrained applications such as the Internet of Things (IoT) and wearable devices.

## 1.3 Product positioning

The ESP32-S3 Mini Module is an ultra-compact embedded development module designed for IoT developers, makers and manufacturers of small electronic devices. Centred around a high-performance microprocessor, the module integrates a wide range of wireless connectivity features and peripheral interfaces. Whilst maintaining an extremely compact form factor, it offers powerful computing capabilities and flexible expandability, making it particularly suitable for space-constrained smart devices and IoT applications.

**Target Audience: IoT application developers, makers and DIY enthusiasts, educational institutions and students, electronics manufacturers, smart home solution providers, and electronics design teams**

## 1.4 Use cases

### 1.4.1 Smart wearables and motion-sensing devices

#### 1. Motion capture equipment

Features: Six-axis IMU (gyroscope + accelerometer) + low power consumption

Description: Designed for developing fitness trackers and posture analysers, this device uses the IMU to accurately capture data such as posture, step count and angular velocity, which is then synchronised via Wi-Fi to a mobile app for statistical analysis; the low-power design ensures long battery life for wearable devices.

### 1.4.2 IoT environmental monitoring

#### 1. Indoor and outdoor environmental monitoring stations

Features: Temperature sensor + Wi-Fi + Peripheral expansion port

Description: Utilises an on-board temperature and humidity module to collect environmental data, which is then uploaded to a cloud platform via Wi-Fi; external gas and light sensors can be connected to expand the scope of monitoring, making it suitable for real-time environmental monitoring in locations such as server rooms, warehouses, greenhouses and buildings.

#### 1.4.3 Smart home

##### 1. Smart home sensor node

Features: Temperature sensor + Wi-Fi + I/O expansion

Description: Designed for indoor home use, it collects temperature data and uploads it to a smart home gateway; it can be connected to relays and lighting modules to enable wireless control of home devices such as lights and sockets.

##### 2. Low-power data logger

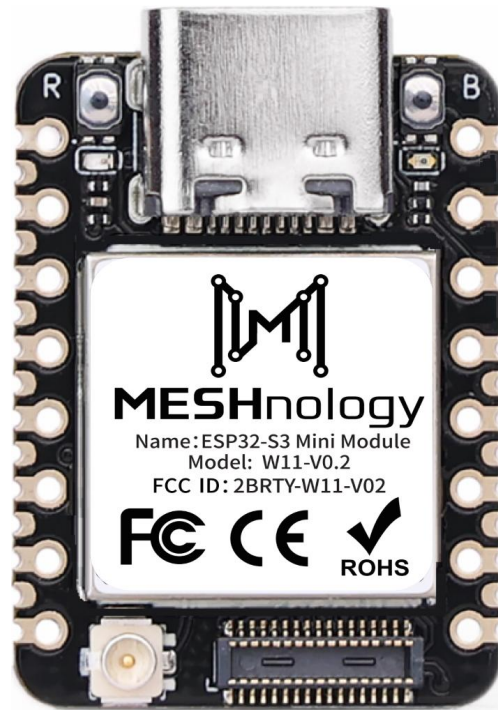
Features: RTC (Real-Time Clock) + various sensors + low-power mode + SD storage

Description: Combines the RTC to accurately stamp data with timestamps, alongside temperature, humidity and attitude sensors, to collect and store data locally over the long term; supports scheduled wake-up and ultra-low-power sleep modes, making it suitable for long-term data logging at remote sites and in unmanned devices, with data transmitted in bulk once connected to the network.

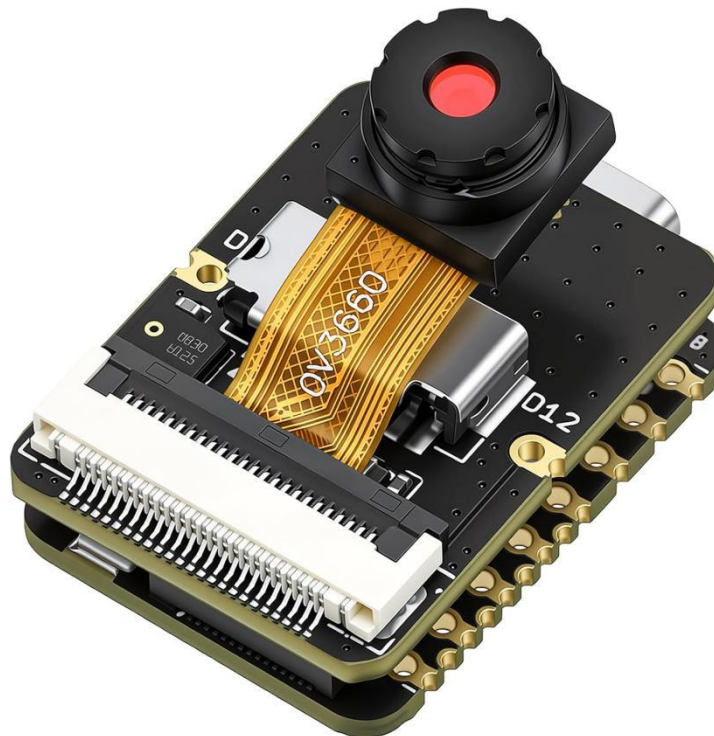
### 1.5 Product highlights

1. **High-performance compact design:** Features an ESP32-S3 dual-core 240MHz processor with NPU, 16MB of on-board Flash and 8MB of PSRAM. Measuring 24.54mm × 17.78mm × 4.50mm, it strikes a balance between computing power and space requirements, making it ideal for wearables, IoT and other applications.
2. **Comprehensive Connectivity and Expansion:** Supports Wi-Fi 802.11b/g/n, up to 32 GPIO channels and multiple high-speed bus interfaces, with on-board temperature and accelerometer sensors, and support for a wide range of peripherals and sensors.
3. **Low power consumption and enhanced security:** Multiple power supply modes + charging protection, with power consumption in deep sleep mode  $\leq 8\mu\text{A}$ ; hardware encryption, secure boot and tamper protection, balancing battery life and data security.
4. **Full-cycle development support:** Compatible with Arduino, ESP-IDF and MicroPython, it offers over 80 example codes, comprehensive hardware documentation and video tutorials, enabling everyone from beginners to professional developers to get started quickly.

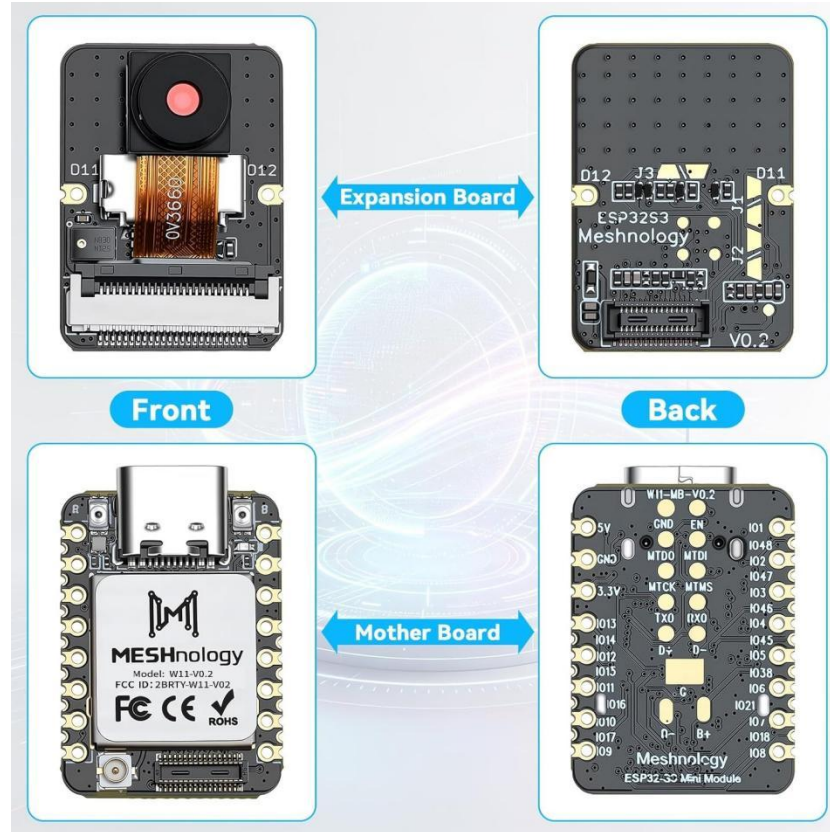
## 1.6 Product prototype



ESP32-S3 Mini Module



Full 3D view of the product



Product Appearance and Structural Diagram

## 2 Product Feature Definition

### 2.1 Product Features

#### 2.1.1 Core processors and computing power

- Equipped with an ESP32S3 dual-core 32-bit LX7 microprocessor, supporting a maximum clock speed of 240 MHz;
- Features 512KB of on-board SRAM and supports hardware-accelerated floating-point operations and vector instruction sets;
- Integrated neural network accelerator (NPU), supporting AI inference at INT8/FP16 precision (e.g. pose recognition, simple image classification);
- Supports coprocessors (enabling peripherals such as RTC, UART and SPI to operate independently, thereby reducing the load on the main core);

#### 2.1.2 Storage Configuration (Enhanced)

- On-board 16MB SPI NOR Flash (compatible with OTA firmware updates, supporting larger application storage);
- Built-in 8MB PSRAM (supports high-resolution image caching and temporary storage of multitasking data, addressing the issue of insufficient memory);
- Supports SPI interface for expanding external storage (compatible with SD cards up to 128GB, meeting the need for local storage of large amounts of data);
- Supports encrypted firmware storage and secure boot (to prevent firmware tampering or reverse engineering);

#### 2.1.3 Wireless connection

- **Wi-Fi Features:**
  - Supports the 802.11b/g/n standard (2.4 GHz band), with a maximum data rate of 150 Mbps;
  - Supports Station, SoftAP and Station+SoftAP hybrid modes, and is compatible with mainstream routers and hotspot devices;

- Supports WPA3/WPA2-PSK security protocols to enhance wireless communication security;

#### 2.1.4 Input/output interfaces (highly scalable)

- Offers up to 32 reusable GPIO pins, all of which support interrupt triggering (rising edge / falling edge / both edges);
- Key interface configurations (all support hardware flow control or high-speed transmission):
  - 20-channel 12-bit ADC (input range 0 – 3.3 V, suitable for analogue sensors such as photodetectors and gas sensors);
  - 3 SPI interfaces (1 high-speed SPI (40 MHz) for PSRAM/Flash, and 2 general-purpose SPI interfaces for peripheral expansion);
  - 2-channel I2C interface (supports 1 MHz high-speed mode, compatible with most I2C sensors and displays);
  - 3 UART interfaces (maximum baud rate 2 Mbps, supports RS485/RS232 expansion, suitable for industrial equipment);
  - 2-channel I2S interface (supports audio input/output; can be connected to a microphone and speakers to enable voice interaction);
  - 1 x USB Type-C port (supporting data transfer (12 Mbps), 5V power supply and firmware flashing);
  - 1-channel lithium battery interface (with charge management circuit, supports 3.7V lithium batteries, adjustable charging current (max. 1A));
- Additional feature interfaces:
  - 2-channel DAC output (8-bit resolution, for analogue signal generation);
  - 8-channel PWM output (supports motor control and LED dimming);
  - On-board user buttons (for reset or triggering custom functions);
  - On-board RGB LED (for status indication, with customisable colours);

#### 2.1.5 Sensor and Hardware Enhancements

- **On-board sensors (standard):**
  - High-precision temperature sensor (accuracy  $\pm 0.1^{\circ}\text{C}$ , measurement range  $-40^{\circ}\text{C}$  to  $125^{\circ}\text{C}$ );
  - High-precision inertial sensors (accelerometer range:  $\pm 2\text{g}$  to  $16\text{g}$ ; gyroscope range:  $\pm 16^{\circ}/\text{s}$  to  $\pm 2048^{\circ}/\text{s}$ );
- **Support for external sensors:**
  - Sensors for temperature and humidity, light, gas, distance, gyroscopes, etc., compatible with I2C/SPI/ADC interfaces;
  - Supports sensor wake-up functionality (wakes the module from deep sleep via sensor signals to reduce power consumption);

#### 2.1.6 Power Management (Low-Power Optimisation)

- Supports multiple power supply modes:
  - Powered via USB Type-C (5V/2A);
  - Powered by a lithium battery (3.7V, supports capacities from 300mAh to 2000mAh);
  - External DC power supply (3.3V–5V, capable of handling a peak current of 2A);
- Charging management functions:
  - Automatically detects charging status (charging / fully charged / fault), with LED indicator;
  - Overcharge / over-discharge / overcurrent / short-circuit protection, ensuring the safety of lithium-ion batteries;
- Low-power mode (suitable for scenarios requiring extended battery life):
  - Active mode: Power consumption  $\leq 120\text{ mA}$  (typical value, under full load);
  - Light sleep mode: Power consumption  $\leq 15\text{ mA}$  (with Wi-Fi standby enabled);
  - Deep sleep mode: Power consumption  $\leq 8\text{ }\mu\text{A}$  (only the RTC and sensors remain active);

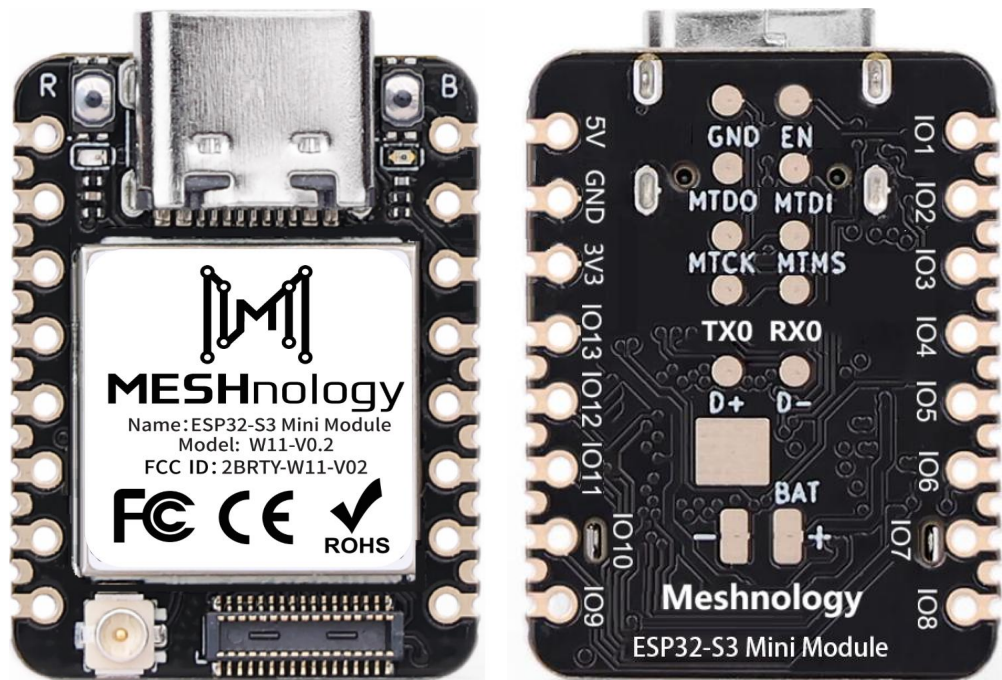
- Sleep mode: Power consumption  $\leq 1 \mu\text{A}$  (only critical wake-up sources, such as GPIO interrupts, are retained);

### 2.1.7 Security features (end-to-end protection)

- Hardware encryption engine (supporting AES-128/256, SHA-256 and RSA-2048 encryption algorithms);
- Secure boot (allows only verified firmware to run, preventing the injection of malicious code);
- Flash memory encryption (encryption of firmware and sensitive data to prevent physical reading);
- Unique Device Identifier (UID), supporting unique device authentication;
- Tamper-proof detection (automatically triggers data erasure if the module is detected to have been disassembled);

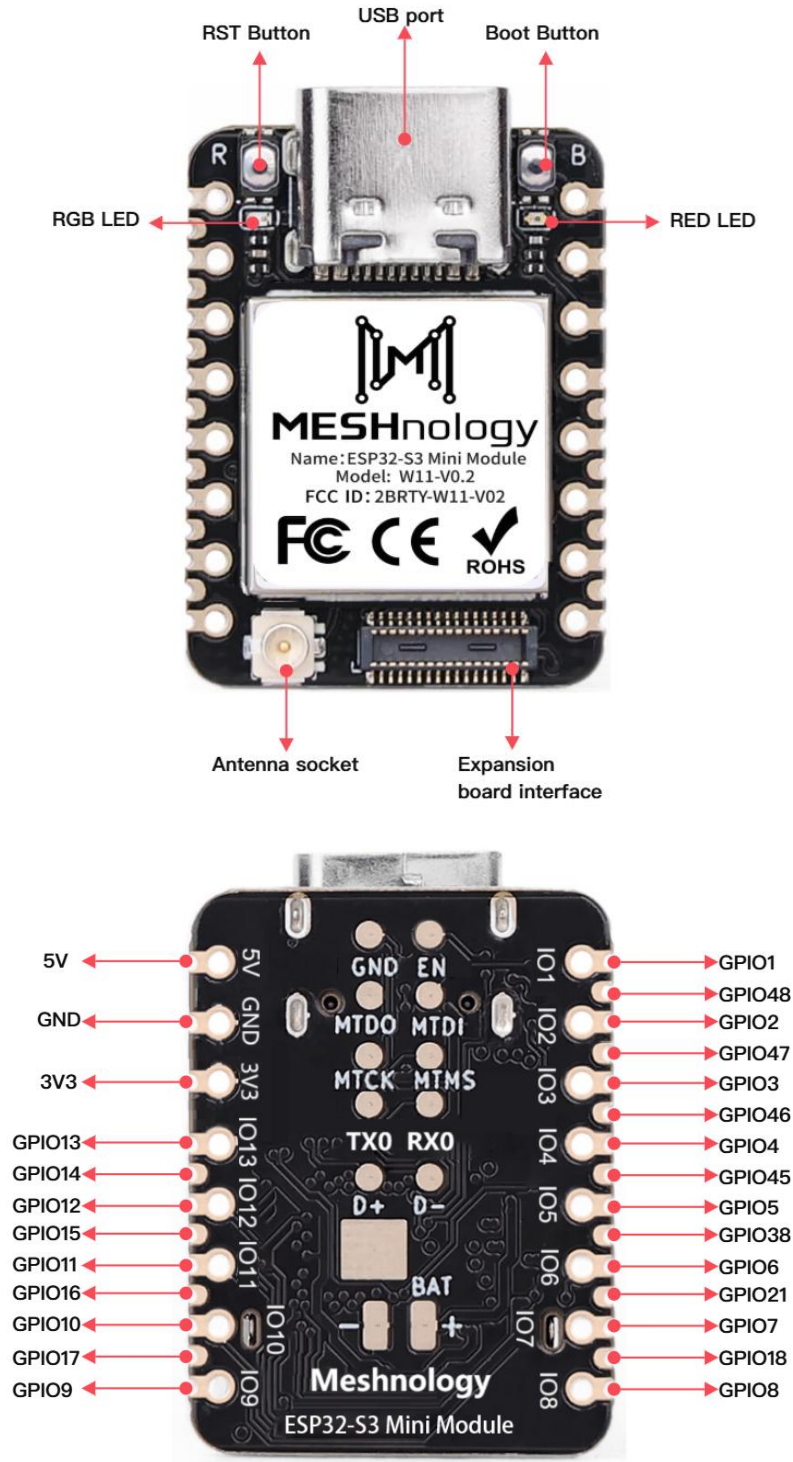
## 2.2 Definition of structural appearance

The product images are shown below:

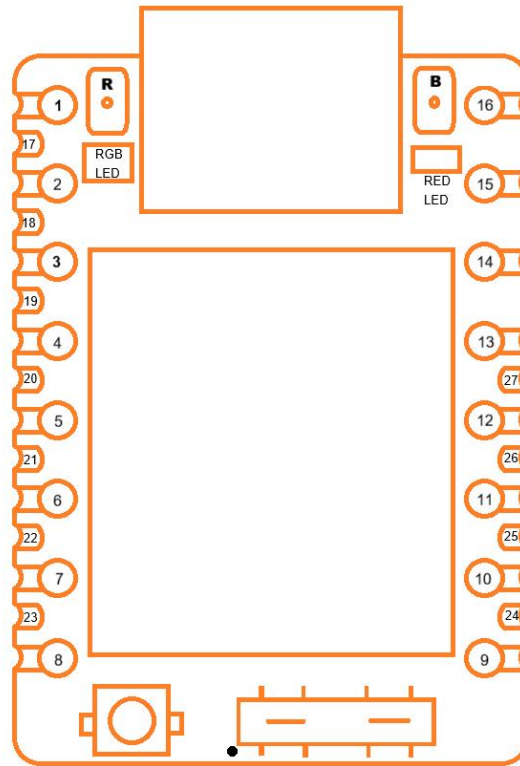


Requirements: The design should be elegant and simple; the PCB should be coated in black, with white silkscreen printing.

## 2.2.1 Hardware Pin Definitions



Front of the module:



1、The pin holes on the left-hand side of the module, from top to bottom

| No. | Name | Type | Function                             |
|-----|------|------|--------------------------------------|
| 1   | IO1  | I/O  | GPIO1, ADC, UART, PWM, I2S, I2C, SPI |
| 2   | IO2  | I/O  | GPIO2, ADC, UART, PWM, I2S, I2C, SPI |
| 3   | IO3  | I/O  | GPIO3, ADC, UART, PWM, I2S, I2C, SPI |
| 4   | IO4  | I/O  | GPIO4, ADC, UART, PWM, I2S, I2C, SPI |
| 5   | IO5  | I/O  | GPIO5, ADC, UART, PWM, I2S, I2C, SPI |
| 6   | IO6  | I/O  | GPIO6, ADC, UART, PWM, I2S, I2C, SPI |
| 7   | IO7  | I/O  | GPIO7, ADC, UART, PWM, I2S, I2C, SPI |
| 8   | IO8  | I/O  | GPIO8, ADC, UART, PWM, I2S, I2C, SPI |

2、The pin holes on the right-hand side of the module, from top to bottom

| No. | Name | Type | Function                                      |
|-----|------|------|-----------------------------------------------|
| 16  | 5V   | P    | Power input or output 5.0V                    |
| 15  | GND  | G    | Ground                                        |
| 14  | 3V3  | P    | Output 3.3V, power supply for external sensor |
| 13  | IO13 | I/O  | GPIO13, ADC, UART, PWM, I2S, I2C, SPI         |
| 12  | IO12 | I/O  | GPIO12, ADC, UART, PWM, I2S, I2C, SPI         |
| 11  | IO11 | I/O  | GPIO11, ADC, UART, PWM, I2S, I2C, SPI         |
| 10  | IO10 | I/O  | GPIO10, ADC, UART, PWM, I2S, I2C, SPI         |
| 9   | IO9  | I/O  | GPIO9, ADC, UART, PWM, I2S, I2C, SPI          |

3、SMA connector sockets on the left-hand side of the module, from top to bottom

| No. | Name | Type | Function |
|-----|------|------|----------|
|-----|------|------|----------|

|    |      |     |                                       |
|----|------|-----|---------------------------------------|
| 17 | IO48 | I/O | GPIO48, UART, PWM, I2S, I2C, SPI      |
| 18 | IO47 | I/O | GPIO47, UART, PWM, I2S, I2C, SPI      |
| 19 | IO46 | I/O | GPIO46, UART, PWM, I2S, I2C, SPI      |
| 20 | IO45 | I/O | GPIO45, UART, PWM, I2S, I2C, SPI      |
| 21 | IO38 | I/O | GPIO38, UART, PWM, I2S, I2C, SPI      |
| 22 | IO21 | I/O | GPIO21, UART, PWM, I2S, I2C, SPI      |
| 23 | IO18 | I/O | GPIO18, ADC, UART, PWM, I2S, I2C, SPI |

4、The SMA connector sockets on the right-hand side of the module, from top to bottom

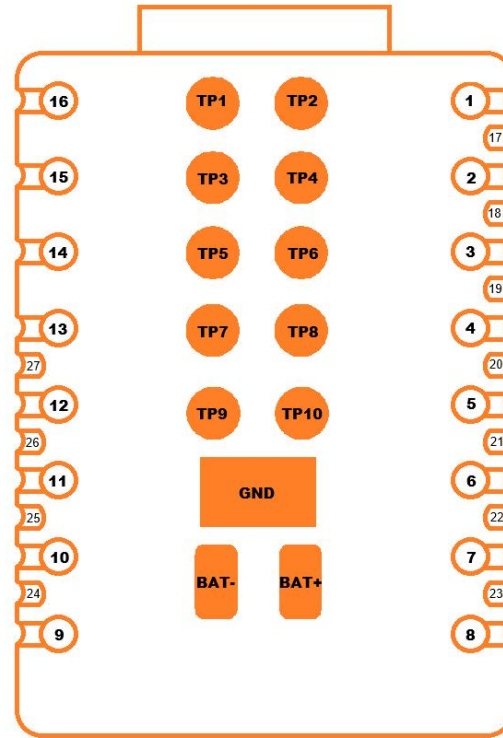
| No. | Name | Type | Function                              |
|-----|------|------|---------------------------------------|
| 27  | IO14 | I/O  | GPIO14, ADC, UART, PWM, I2S, I2C, SPI |
| 26  | IO15 | I/O  | GPIO15, ADC, UART, PWM, I2S, I2C, SPI |
| 25  | IO16 | I/O  | GPIO16, ADC, UART, PWM, I2S, I2C, SPI |
| 24  | IO17 | I/O  | GPIO17, ADC, UART, PWM, I2S, I2C, SPI |

5、DF40C-30-pin socket, from the bottom-left pin to the top-left corner

| No. | Name | Type | Function                                          |
|-----|------|------|---------------------------------------------------|
| 1   | NC   | /    | Reserved                                          |
| 2   | IO3  | I/O  | GPIO3, ADC, UART, PWM, I2S, I2C, SPI              |
| 3   | IO7  | I/O  | GPIO7, ADC, UART, PWM, I2S, I2C, SPI              |
| 4   | IO8  | I/O  | GPIO8, ADC, UART, PWM, I2S, I2C, SPI              |
| 5   | IO9  | I/O  | GPIO9, ADC, UART, PWM, I2S, I2C, SPI              |
| 6   | IO10 | I/O  | GPIO10, ADC, UART, PWM, I2S, I2C, SPI             |
| 7   | IO11 | I/O  | GPIO11, ADC, UART, PWM, I2S, I2C, SPI             |
| 8   | IO12 | I/O  | GPIO12, ADC, UART, PWM, I2S, I2C, SPI             |
| 9   | IO13 | I/O  | GPIO13, ADC, UART, PWM, I2S, I2C, SPI             |
| 10  | IO14 | I/O  | GPIO14, ADC, UART, PWM, I2S, I2C, SPI             |
| 11  | IO15 | I/O  | GPIO15, ADC, UART, PWM, I2S, I2C, SPI             |
| 12  | IO16 | I/O  | GPIO16, ADC, UART, PWM, I2S, I2C, SPI             |
| 13  | IO17 | I/O  | GPIO17, ADC, UART, PWM, I2S, I2C, SPI             |
| 14  | IO18 | I/O  | GPIO18, ADC, UART, PWM, I2S, I2C, SPI             |
| 15  | VIN  | P    | Power input for module, Voltage range is 3.7~5.0V |
| 16  | IO21 | I/O  | GPIO21, UART, PWM, I2S, I2C, SPI                  |
| 17  | GND  | G    | Ground                                            |
| 18  | 3V3  | P    | Output 3.3V, power supply for external sensor     |
| 19  | 3V3  | P    | Output 3.3V, power supply for external sensor     |
| 20  | NC   | /    | Reserved                                          |
| 21  | IO48 | I/O  | GPIO48, UART, PWM, I2S, I2C, SPI                  |
| 22  | IO47 | I/O  | GPIO47, UART, PWM, I2S, I2C, SPI                  |
| 23  | IO38 | I/O  | GPIO38, UART, PWM, I2S, I2C, SPI                  |
| 24  | IO39 | I/O  | GPIO39, UART, PWM, I2S, I2C, SPI                  |
| 25  | IO40 | I/O  | GPIO40, UART, PWM, I2S, I2C, SPI                  |
| 26  | IO41 | I/O  | GPIO41, UART, PWM, I2S, I2C, SPI                  |
| 27  | IO42 | I/O  | GPIO42, UART, PWM, I2S, I2C, SPI                  |
| 28  | GND  | G    | Ground                                            |
| 29  | GND  | G    | Ground                                            |
| 30  | VIN  | P    | Power input for module, Voltage range is 3.7~5.0V |

Schematic diagram of the DF40C-30-pin socket connector

the back of the module :



1、The TP pads in the middle of the module, from top to bottom

| No.  | Name | Type | Function                                        |
|------|------|------|-------------------------------------------------|
| TP1  | GND  | I/O  | Ground                                          |
| TP2  | EN   | I/O  | CHIP_PU, Connect to RST switch                  |
| TP3  | MTDO | I/O  | MTDO, GPIO40, UART, PWM, I2S, I2C, SPI          |
| TP4  | MTDI | I/O  | MTDI, GPIO41, UART, PWM, I2S, I2C, SPI          |
| TP5  | MTCK | I/O  | MTCK, GPIO39, UART, PWM, I2S, I2C, SPI          |
| TP6  | MTMS | I/O  | MTMS, GPIO42, UART, PWM, I2S, I2C, SPI          |
| TP7  | TX0  | I/O  | U0TXD, GPIO43, UART, PWM, I2S, I2C, SPI         |
| TP8  | RX0  | I/O  | U0RXD, GPIO44, UART, PWM, I2S, I2C, SPI         |
| TP9  | D+   | I/O  | USB_D+, GPIO20, ADC, UART, PWM, I2S, I2C, SPI   |
| TP10 | D-   | I/O  | USB_D-, GPIO19, ADC, UART, PWM, I2S, I2C, SPI   |
| GND  | GND  | G    | Ground                                          |
| BAT- | BAT- | P    | Input 3.7~4.2V, Battery negative terminal input |
| BAT+ | BAT+ | P    | Input 3.7~4.2V, Battery positive terminal input |

### 2.2.2 Definition of hardware peripherals

- 1、RGB LED interface: connected via GPIO48;
- 2、Sensors: The temperature sensor and inertial sensor are connected via GPIO4 (SCL), GPIO5 (SDA) and GPIO6 (INT);
- 3、Battery voltage: Battery voltage is monitored using GPIO2 (ADC);

### 2.2.3 Module label definition

Explanation of label fields:

| Field              | Description                                                 |
|--------------------|-------------------------------------------------------------|
| Model              | ESP32-S3 Mini Module                                        |
| FCC-ID             | Unique certification number for radio transmission products |
| Certification Logo | CE, RoHS and FCC compliance marks                           |

Labelling requirements:

Screen-printed colour: white (to match the white screen-printing process on the board);

Font: Industrial-standard sans-serif font, clear and durable, capable of withstanding the high temperatures of reflow soldering;

Placement: Avoid antenna areas, RF circuits and metal pads to prevent interference with RF performance.

#### 2.2.4 Antenna Design and Cabling Specifications

This module is equipped with a 2.4GHz Wi-Fi antenna; the following rules must be observed when integrating it onto the main PCB:

##### 1. Antenna clearance requirements

Do not lay copper wiring or install power or signal cables within the antenna area; do not place metal components such as metal casings, screws or shielding covers within 15 mm of the antenna or directly beneath it.

##### 2. RF routing requirements

The RF circuits are designed strictly to a 50Ω impedance, with trace lengths kept as short as possible to minimise signal attenuation.

##### 3. Interference avoidance

Place components that generate strong interference—such as the main controller, power supply and motors—well away from the antenna area, and add filter components to the power supply circuit to suppress noise.

##### 4. On-board antenna specifications

FPC antenna, gain 4.07 dBi; external antenna uses a standard IPEX 1.0 connector.

## 2.3 List of Certifications

| Certification category | Reason for use                                   | Notes                                                              |
|------------------------|--------------------------------------------------|--------------------------------------------------------------------|
| FCC/IC/CE-RED          | WiFi radio frequency transmission                | Regulations on wireless equipment to ensure radio frequency safety |
| EMC/EMI                | Digital interfaces, radio frequency interference | Electromagnetic Compatibility and Interference Suppression         |

|            |                     |                                                                     |
|------------|---------------------|---------------------------------------------------------------------|
| RoHS/REACH | PCBs and components | Restrictions on hazardous substances and environmental requirements |
|------------|---------------------|---------------------------------------------------------------------|

## 2.4 Compliance with FCC Regulations and Operating Conditions

### 1. Applicable legislation

This RF module has been certified in accordance with FCC Part 15 Subpart C; the module does not include certification under FCC Part 15 Subpart B (digital spurious emissions), for which the terminal equipment manufacturer is responsible for testing compliance.

### 2. Limitations of the groundwork

Rated supply voltage: 3.3V to 5.0V DC. Do not exceed the voltage rating or connect in reverse.

Operating temperature: -40°C to 85°C; operating outside this range may affect RF performance and service life.

Antenna restrictions: Only the antenna certified for use with the module may be used; unauthorised replacement or modification of the RF antenna or cabling is prohibited.

### 3. Requirements for multi-RF co-location

If the terminal integrates other RF transmission modules, such as LoRa and cellular, and these operate simultaneously, the original equipment manufacturer must conduct assessments and additional testing in accordance with the FCC's multi-transmitter rules and, where necessary, submit a Class II licence amendment.

### 4. Module type

This product is a standard, full-featured RF module, not a limited module.

## 2.5 Requirements for radiofrequency exposure

- Minimum safe distance between the module antenna and the human body:  $\geq 20$  cm;
- When the equipment is in operation, do not remain in close contact with the antenna area for prolonged periods.

## 2.6 Antenna Specifications and Usage Restrictions

1. Antenna type: 4.07 dBi FPC antenna + external IPEX 1.0 antenna connector;
2. Permitted antenna types: monopole antennas, PIFA antennas, dipole antennas;
3. The use of uncertified third-party antennas and non-standard RF connectors is prohibited;
4. The antenna gain and impedance must not exceed the module's certified parameters; otherwise, the certification will be invalidated.

## 2.7 Labelling compliance requirements

Host product manufacturers need to provide a physical or e-label stating "Contains Transmitter Module FCC ID: 2BRTY-W11-V02" with their finished product.

## 2.8 Test modes and additional test requirements for the complete unit

### 1. WIFI

Operation Frequency: 2412~2462MHz

Number of Channel: 11

Modulation: DSSS

Host manufacturer must perform test of radiated & conducted emission and spurious emission, etc according to the actual test modes for a stand-alone modular transmitter in a host, as well as for multiple simultaneously transmitting modules or other transmitters in a host product. Only when all the test results of test modes comply with FCC requirements, then the end product can be sold legally.

## 2.9 EMI Design Recommendations

- Recommendations for EMC design of the complete unit:

Power chokes and high-frequency filter capacitors have been added around the module to suppress power supply noise;

The host ensures a completely flat surface to minimise ground loop interference;

High-speed signal lines and clock lines should be kept well away from the module's RF section and the antenna;

Ensure that multiple RF modules are laid out with adequate shielding and isolation to prevent intermodulation interference;

During EMC testing of the complete unit, the module must operate under typical emission conditions.

## 2.10 Product Change Management Policy

- Non-RF changes (software features, general I/O, peripheral circuits of the module): No FCC filing or approval required, only internal design assessment by the module grantee.
- RF-related changes (antennas, RF matching circuits, transmit output power, operating frequency bands, RF firmware): Only the module manufacturer (FCC Grantee) is authorized to file a Class II Permissive Change via TCB before mass production. OEM integrators are not allowed to submit permissive change applications.
- Major hardware redesign (replacement of the main MCU or RF chip on the module): The module manufacturer shall complete a full new FCC certification procedure and obtain a new FCC grant prior to release.
- OEM/host product manufacturers are strictly prohibited from making any unauthorized modifications to the module's RF components, antennas or RF PCB traces. Any RF-related alteration to the module will invalidate the original modular FCC authorization, and the host product shall undergo full independent FCC certification separately.

## 2.11 Product Specifications

| Category                           | Parameter items                 | Specific technical specifications                                                                                                                                                                                                                                                                  |
|------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core processor</b>              | Chip model                      | ESP32-S3 (dual-core 32-bit LX7 microprocessor)                                                                                                                                                                                                                                                     |
|                                    | Main frequency                  | Highest 240MHz                                                                                                                                                                                                                                                                                     |
|                                    | On-board SRAM                   | 512KB                                                                                                                                                                                                                                                                                              |
|                                    | Built-in PSRAM                  | 8MB (Supports high-resolution image caching and multi-task data storage)                                                                                                                                                                                                                           |
|                                    | Key features                    | Hardware floating-point operations, vector instruction sets, neural network accelerators (NPUs, supporting AI inference at INT8/FP16 precision)                                                                                                                                                    |
| <b>Storage configuration</b>       | On-board Flash                  | 16MB SPI NOR Flash (Supports OTA firmware updates and firmware encryption)                                                                                                                                                                                                                         |
|                                    | Support for expandable storage  | SPI interface expansion, compatible with SD cards up to 128GB                                                                                                                                                                                                                                      |
| <b>Wireless connection</b>         | Wi-Fi Standard / Band           | 802.11b/g/n, 2.4GHz frequency band                                                                                                                                                                                                                                                                 |
|                                    | Wi-Fi Data rate / Distance      | Up to 150 Mbps, indoor range $\geq 50$ m, outdoor range $\geq 100$ m (packet loss rate $\leq 1\%$ at 100 m)                                                                                                                                                                                        |
|                                    | Wi-Fi Modus / Security protocol | Station/SoftAP/ Hybrid mode; Support WPA3/WPA2-PSK                                                                                                                                                                                                                                                 |
| <b>Input and output interfaces</b> | GPIO Number of pins             | Up to 32 reusable GPIO pins (all supporting interrupt triggering: rising edge / falling edge / both edges)                                                                                                                                                                                         |
|                                    | ADC Passage                     | 20 channels, 12-bit resolution, input range 0–3.3 V                                                                                                                                                                                                                                                |
|                                    | SPI Interface                   | 3 channels (1 high-speed SPI channel at 40 MHz for PSRAM/Flash, and 2 general-purpose SPI channels for expansion peripherals)                                                                                                                                                                      |
|                                    | UART Interface                  | 2-channel, supports 1 MHz high-speed mode                                                                                                                                                                                                                                                          |
|                                    | UART Interface                  | 3 channels, maximum baud rate 2 Mbps (supports hardware flow control)                                                                                                                                                                                                                              |
|                                    | Other key interfaces            | 2 x I2S (audio input/output), 2 x DAC (8-bit resolution), 8 x PWM, 1 x USB Type-C (data/power/programming)                                                                                                                                                                                         |
|                                    | Additional feature interfaces   | Lithium battery connector (3.7V), user buttons (reset / custom), RGB LED (status indicator)                                                                                                                                                                                                        |
| <b>Sensor Configuration</b>        | On-board sensors (standard)     | 1. Temperature sensor: Accuracy $\pm 0.1^{\circ}\text{C}$ , measurement range $-40^{\circ}\text{C}$ to $125^{\circ}\text{C}$ ;<br>2. Inertial sensors: Accelerometer range $\pm 2\text{g}$ to $16\text{g}$ ; gyroscope range $\pm 16^{\circ}/\text{s}$ to $\pm 2048^{\circ}/\text{s}$ (selectable) |
|                                    | Support for external sensors    | Sensors such as temperature and humidity, light, gas, distance and gyroscope sensors compatible with I2C/SPI/ADC interfaces (supports sensor wake-up)                                                                                                                                              |

|                                                   |                                                                                                                                                                                                                           |                                                                                                                                                          |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Power management</b>                           | Power supply method                                                                                                                                                                                                       | 1. USB Type-C (5V/1A) ;<br>2. lithium-ion battery (3.7V, 300mAh~2000mAh) ;<br>3. External DC (3.3V~5V/1A peak)                                           |
|                                                   | Charging management                                                                                                                                                                                                       | Adjustable charging current (up to 1A) with LED charging status indicator; features overcharge, over-discharge, overcurrent and short-circuit protection |
|                                                   | Power consumption (typical)                                                                                                                                                                                               | Active mode $\leq 120$ mA; light sleep mode $\leq 15$ mA; deep sleep mode $\leq 8$ $\mu$ A; sleep mode $\leq 1$ $\mu$ A                                  |
| <b>Physical and Environmental Characteristics</b> | Dimensions                                                                                                                                                                                                                | 24.54mm $\times$ 17.78mm $\times$ 4.50mm                                                                                                                 |
|                                                   | Weight                                                                                                                                                                                                                    | 2.44g (Single ESP32-S3 module)                                                                                                                           |
| <b>Physical and Environmental Characteristics</b> | PCB Specifications                                                                                                                                                                                                        | 1.6 mm thick FR-4 board, gold-plated finish                                                                                                              |
|                                                   | Operating temperature range                                                                                                                                                                                               | -40°C to 85°C (industrial grade)                                                                                                                         |
|                                                   | Storage temperature range                                                                                                                                                                                                 | -55°C~125°C                                                                                                                                              |
|                                                   | Operating humidity                                                                                                                                                                                                        | 5%~95% RH (No condensation)                                                                                                                              |
| <b>Reliability</b>                                | Mean Time Between Failures (MTBF)                                                                                                                                                                                         | $\geq 20,000$ hour                                                                                                                                       |
|                                                   | Electromagnetic compatibility (EMC)                                                                                                                                                                                       | Complies with CE EN 300 328 and FCC Part 15B standards                                                                                                   |
|                                                   | Electrostatic protection (ESD)                                                                                                                                                                                            | Contact discharge $\pm 8$ kV, air discharge $\pm 15$ kV (IEC 61000-4-2 standard)                                                                         |
|                                                   | IO output current limit                                                                                                                                                                                                   | Per pin $\leq 20$ mA; total I/O current $\leq 100$ mA                                                                                                    |
|                                                   | Vibration-resistant / Shock-resistant                                                                                                                                                                                     | Vibration resistance: 10–2000 Hz, 10 g acceleration (IEC 60068-2-6); Shock resistance: 1000 g, 0.5 ms half-sine wave (IEC 60068-2-27)                    |
| <b>Notes</b>                                      | 1. RF specifications comply with FCC Part 15C and CE-RED standards; 2. The digital circuitry of the entire unit must undergo FCC Part 15B testing; 3. ESD protection: contact $\pm 8$ kV, air $\pm 15$ kV (IEC 61000-4-2) |                                                                                                                                                          |

## 2.12 OEM Integration Compliance Requirements

### 2.12.1 List of FCC regulations applicable

This wireless module has obtained only FCC Part 15 Subpart C transmission RF certification, covering the 2.4 GHz Wi-Fi 802.11b/g/n frequency band, with specified transmission power limits, stray radiation levels, and fundamental operating frequency controls.;

This module certification does not cover the FCC Part 15 Subpart B unintentional radiation standards. The original equipment manufacturer (OEM) is fully responsible for conducting 15B conduction/radiation

disturbance tests and ensuring compliance for all digital circuits, motherboard components, and peripheral devices in the final product.;

OEM manufacturers may not rely solely on this module's FCC certification exemption to bypass the full Unit 15B test; they must complete the entire Part 15B electromagnetic compatibility test before shipping the final product..

### 2.12.2 Device Usage Restrictions and Operating Conditions

Workplace restrictions: The module supports both indoor and outdoor deployment but cannot be installed on aircraft or airborne devices;

Antenna gain restrictions: Only the certified 4.07 dBi FPC onboard antenna and compliant single-pole, PIFA, or dipole external antennas are permitted; antennas with gain>4.07 dBi are prohibited.;

Cable loss compensation: When using an external antenna extension cable, the OEM must adjust the module's transmission power based on the RF cable attenuation to ensure the overall radiation power does not exceed certified limits.;

Power Supply Restrictions: The module requires a DC supply of 3.3 V to 5.0 V only. Overvoltage or reversed polarity will cause RF exceedance and result in failure certification.;

Operating Temperature: The entire system must maintain the module's ambient temperature within -40°C to 85°C; exceeding this range will cause RF performance drift and violate compliance requirements.

### 2.12.3 Finite Module Description

This W11 (ESP32-S3 Mini Module) is a standard full module, not a limited version. This W11 (ESP32-S3 Mini Module) is a standard full module, not a limited version.:

Host manufacturers do not need to submit host design reviews to module manufacturers in advance; they can integrate directly.;

To ensure compatibility and coexistence of the module with other RF transmitters, OEMs must conduct independent multi-transmitter coexistence evaluations without requiring module manufacturers to follow the host device approval process.;

The module manufacturer must initiate a Class II license change only when the host usage, antenna configuration, or RF layout significantly deviates from the parameters specified in this specification.

### 2.12.4 Wiring Antenna Specifications

This module supports two antenna configurations. The OEM host PCB design must strictly adhere to the following parameters; any trace deviation must be reported to the module manufacturer and a Class II modification request submitted.:

Tolerance specifications for the wiring layout of the onboard FPC antenna

Wiring Impedance: Constant 50  $\Omega$  differential RF trace;

Media dielectric constant: FR4 board 4.2–4.4;

Wiring width/thickness: board thickness 1.6 mm FR4, RF line width 1.8 mm, copper foil thickness 1 oz;

Isolation Zone: Copper traces, power cables, signal cables, metal components, and shielding covers are prohibited within 15 mm of the antenna area.;

External IPEX 1.0 Antenna Interface Specifications;

Only standard IPEX 1.0 RF connectors are allowed; alternative specifications are prohibited;

RF trace optimization to minimize signal attenuation;

Verification Requirements

The OEM host must complete factory tests for antenna gain and standing wave ratio.;

Changes to antenna layout, trace dimensions, and board materials constitute RF modifications that must be submitted as FCC Class II permitted changes.

#### 2.12.5 RF (Radio Frequency) Exposure Compliance Requirements

Document requirements are divided into two categories: specifications for OEM integrators and mandatory wording for end-user manuals.

Hard requirements for OEM integrated design

The overall structure of the device must ensure that the minimum safe distance between the antenna radiator and the human body is  $\geq 20$  cm.

It cannot be designed as a close-fitting portable device (within 20 cm of the body). If intended for wearable devices, OEMs must independently conduct SAR testing and obtain additional certification.

In multi-module coexistence scenarios, the MPE radio frequency exposure limits must be re-evaluated for multi-transmitter antenna interference mitigation.

Radio Frequency Exposure Statement: This device complies with FCC limits for radio frequency radiation in uncontrolled environments. During operation, maintain a minimum safety distance of 20 cm between the antenna and the human body. Close or prolonged exposure to the antenna area is prohibited.

#### 2.12.6 Antenna Compliance List and Connector Specifications

List of Certified Antenna Types (Only the following are available)

The onboard FPC features an integrated antenna with a gain of 4.07 dBi;

External single-pole antenna (max 4.07 dBi);

PIFA planar inverted-F antenna (max 4.07 dBi);

Dipole Antenna (Max 4.07 dBi);

Hard requirements for RF connectors

Only IPEX 1.0 standard RF connectors are allowed; custom non-standard connectors are prohibited;

Host PCB RF pad matching module IPEX pin definition: Pad layout cannot be modified;

Prohibited Actions: Do not replace high-gain antennas, modify RF connectors, or connect multiple antennas in parallel without authorization.

#### 2.12.7 Compliance of the entire device label with OEM requirements

Module Body Label: Permanent screen-printed identification for module model W11-V0.2; FCC ID: 2BRTY-W11-V02; Resistant to high temperatures during reflow soldering.;

Mandatory labeling for OEM finished products (choose between permanent physical label or electronic e-label; refer to KDB784748)

The entire unit must be printed: Contains FCC ID: 2BRTY-W11-V02

Label placement requirements: Labels must be clearly visible, not obstructed or worn. They must not be placed in the antenna's RF area.;

Electronic Tag Solution: If the entire device uses an e-label, the user manual must specify the e-label retrieval path in compliance with FCC e-label specifications.

#### 2.12.8 Overall Machine Testing Mode and Additional Testing Requirements

The module includes an independent RF testing mode.

Supports enabling a dedicated Wi-Fi transmit/receive channel for production line RF calibration and stray radiation testing;

Supports two testing scenarios: independent operation of a single module and synchronous transmission of multiple radio frequencies;

OEM requires mandatory additional testing items for the entire device.

1) The entire device, after being equipped with the module, passes the complete FCC Part 15 Subpart B unintentional radiation test.;

2) When multiple RF modules coexist, conduct a joint assessment of coexistence-related EMI and RF exposure in accordance with the FCC multi-transmitter procedure.;

3) Re-measurement of RF transmission power and stray radiation under various temperature conditions (high and low);

OEM test record retention: The complete test report must be archived for TCB verification.

#### 2.12.9 Part15 Subpart B disclaimer

OEM manufacturers must include the following exemption statement in the end-user manual.:

This W11 (ESP32-S3 Mini Module) is certified only for RF emission compliance under FCC Part 15 Subpart C; the module certification does not cover unintentional radiation compliance of the entire device's digital circuits. When integrating this module into end products, manufacturers must independently complete all electromagnetic compatibility tests specified in FCC Part 15 Subpart B to ensure full compliance with relevant limit requirements. Even if the module comes with basic EMI test data, the entire device must still undergo retesting after full power-on operation with the module installed.

#### 2.12.10 EMI (Electromagnetic Interference) Integrated Design Guidelines

OEM Host Design Specifications Based on Best Practices for KDB996369 D04 Module Integration Guidelines:

The module power circuit incorporates a magnetic bead and a high-frequency 0402 filter capacitor to suppress power noise coupling into the RF circuit;

The main control chip, motor, high-speed clock, and switching power supply must be kept more than 15 mm away from the antenna;

Add grounding isolation slots and shielding barriers between multiple RF modules to prevent mutual interference;

The overall EMC testing of the device must cover both full-load transmission conditions and multi-RF synchronous operation conditions;

If stray emissions exceed the specified limits during whole-unit testing, the OEM must adjust the layout and add filtering components; modifying the module's RF hardware is strictly prohibited.

#### 2.12.11 Product Change Management and License Change Process

Only the module certification holder (our company) is authorized to submit a Permissive Change to the FCC; OEM manufacturers are prohibited from making unauthorized modifications to the module's RF-related hardware:

Non-RF category changes (software functions, standard GPIOs, peripherals, structural housing): No FCC modification required;

RF-related changes (antenna model, routing, transmission power, RF firmware, RF matching circuit): OEMs must contact us to initiate a Class II license change;

Replace all RF antennas of different types and modify the PCB RF traces: Bulk shipment is prohibited before completing these changes;

Contact Information: OEM integrators may contact our Hardware Compliance Team for assistance with change submissions.

## 3 Compliance Appendix Statement

### 3.1 FCC Official Compliance Statement

English (For overseas products)

This device complies with Part 15 of the FCC Rules.

Operation is subject to the following two conditions:

- This device may not cause harmful interference.
- This device must accept any interference received, including interference that may cause undesired operation.
- Any changes or modifications not expressly approved by the responsible party will void the user's authority to operate the equipment.
- RF Exposure Statement
- This equipment complies with FCC RF exposure limits for uncontrolled environments. Maintain a minimum separation distance of 20 cm between the antenna and human body during operation.

### 3.2 CE & RoHS & WEEE Declaration

- This product complies with the EU Radio Equipment Directive 2014/53/EU and the RoHS 2.0 Directive on the Restriction of Hazardous Substances;
- The 2.4 GHz band may be used throughout the European Union, whilst the 5150–5350 MHz band is restricted to indoor use only;
- In accordance with the EU WEEE Directive (2012/19/EU) on waste electrical and electronic equipment, this item must not be disposed of as ordinary household waste; please take it to a designated recycling centre.

## 4 About Us

We are a high-tech enterprise specialising in the research, development, production and sale of core IoT modules. With many years of experience in the ESP series module sector, we are committed to providing highly reliable, high-performance IoT hardware solutions to customers worldwide.

### 4.1 Key strengths

Technical Capabilities: We have a dedicated hardware R&D team that has extensively optimised the Espressif ESP32-S3 chip technology. We possess end-to-end development capabilities, ranging from

module design to firmware optimisation, and our products undergo rigorous reliability testing and compatibility verification.

**Quality Assurance:** Manufactured to industrial-grade standards using high-quality components and precision manufacturing processes, all modules undergo high- and low-temperature testing, ageing tests and RF performance calibration to ensure stable operation in a wide range of complex environments.

**Support:** We provide comprehensive technical documentation, sample code and one-to-one technical consultation services. We support OEM/ODM customisation requirements and respond swiftly to customers' product iteration and feature adaptation needs.

## **4.2 Areas of application**

Our products are widely used in smart home, industrial automation, smart agriculture, healthcare, wearable devices and IoT gateways, providing customers with one-stop 'hardware + software' IoT solutions to support the digital transformation and upgrading of industries.

## FCC warning:

Any Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Note: This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with minimum distance 20cm between the radiator & your body.